Module 3 slides: Intravenous Infusions (worked examples)

Module 3 - Intravenous Infusions · Lecture slides, worked examples · 2 Sep 2026

Name _____ Date _____

Example 1 Theophylline is given by continuous intravenous infusion at 50 mg/hr. Its half-life is 6 hours and its volume of distribution is 30 L. What steady-state plasma concentration is expected?

Example 2 The same drug has a half-life of 6 hours and a volume of distribution of 30 L. What infusion rate would achieve a steady-state concentration of 20 mg/L?

A patient received an intravenous infusion of 150 mg over a period of 6 hours. The drug has an elimination rate constant of 0.231 hr^{-1} and an apparent volume of

distribution of 15 L.

Example 6 What is the plasma concentration at the end of the infusion?

Example 6 What is the concentration of the drug in the body 3 hours after cessation of the infusion?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Example 1 Her key

Answer: 14.43 mg/L

WHAT THE STEM GIVES

R	50 mg/hr	the numerator of $\frac{R}{Cl}$
t½	6 hours	gives $k = \frac{0.693}{6} = 0.1155\ hr^{-1}$
V_D	30 L	with k gives $Cl = 0.1155 \times 30 = 3.465$ L/hr, the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{ss} = \frac{R}{Cl}$, and clearance is itself $k \cdot V_D$, so the half-life has to become a rate constant before the plateau can be computed

At steady state rate in equals rate out, and rate out is clearance times concentration, so the plateau is infusion rate divided by clearance. The stem gives a half-life and a volume, not a clearance, so two conversions come before the division.

ALGEBRA

$$k = \frac{0.693}{6\ hr} = 0.1155\ hr^{-1}$$

The steady-state concentration needs a clearance, and clearance is the elimination rate constant multiplied by the volume of distribution, so the half-life has to be converted to a rate constant first. Dividing 0.693 by a time in hours leaves reciprocal hours, which is what a first-order rate constant must be.

ALGEBRA

$$Cl = k \cdot V_D = (0.1155\ hr^{-1})(30\ L) = 3.465\ L/hr$$

Clearance is the volume of plasma cleared of drug per unit time, and it is the product of the fractional rate of removal and the volume that drug occupies. Reciprocal hours multiplied by litres leaves litres per hour, the unit a clearance must carry.

ALGEBRA

$$C_{ss} = \frac{R}{Cl} = \frac{50\ mg/hr}{3.465\ L/hr} = 14.43\ mg/L$$

At steady state the rate in equals the rate out, and the rate out is clearance multiplied by concentration, so the concentration is the infusion rate divided by clearance. Milligrams per hour divided by litres per hour cancels the hours and leaves milligrams per litre.

ROUND

$C_{ss} = 14.43$ mg/L

Report 14.43 mg/L, or about 14 mg/L. The two decimal places follow the four-figure rate constant used throughout.

DOES THE NUMBER MAKE SENSE?

> k is 0.693 over 6, 0.1155 hr^{-1}, so Cl is 0.1155 × 30 = 3.465 L/hr; 50 mg/hr spread over 3.465 L/hr is a little over 14 mg/L, and 14.43 fits.

Example 2 Her key

Answer: 69.3 mg/hr

WHAT THE STEM GIVES

t½	6 hours	gives $k = \frac{0.693}{6} = 0.1155\ hr^{-1}$
V_D	30 L	with k gives Cl = 0.1155 × 30 = 3.465 L/hr
C_{ss}	20 mg/L	the target, multiplied by Cl

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = C_{ss} \cdot Cl = C_{ss} \cdot k \cdot V_D$$

Rearranging C_{ss} equal to R over clearance puts the rate on its own, so the wanted concentration is multiplied rather than divided by clearance. Writing clearance as k times V_D lets the two parameters given in the stem be used directly.

ALGEBRA

$$k = \frac{0.693}{6\ hr} = 0.1155\ hr^{-1}$$

The relation between rate and steady-state concentration runs through clearance, and clearance needs the rate constant. Converting the half-life first keeps every later line in reciprocal hours.

ALGEBRA

$$R = (20\ mg/L)(0.1155\ hr^{-1})(30\ L) = 69.3\ mg/hr$$

Milligrams per litre multiplied by reciprocal hours multiplied by litres cancels the litres and leaves milligrams per hour, which is the unit an infusion rate must have. The arithmetic is 20 times 3.465.

ROUND

$$R = 69.3\ mg/hr$$

Report 69.3 mg/hr. One decimal place is what the four-figure rate constant supports. As a check, 69.3 divided by the clearance of 3.465 L/hr returns the 20 mg/L target.

> **DOES THE NUMBER MAKE SENSE?**
> Cl is k × V_D = 0.1155 × 30, about three and a half L/hr, and the rate has to replace what is cleared: 20 mg/L × Cl, about 69.3 mg/hr. A higher target needs a proportionally higher rate.

Example 6 Her key

Answer: 5.41 mg/L

WHAT THE STEM GIVES

dose	150 mg	over the duration gives $R = \frac{150}{6} = 25\ mg/hr$

duration	6 hours	the denominator of R, and the t in the exponent
k	0.231 hr^{-1}	with V_D gives Cl = 3.465 L/hr; the exponent kt = 0.231 × 6
V_D	15 L	one factor of $Cl = k \cdot V_D$

THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = \frac{R}{Cl}(1 - e^{-kt})$ at the end of the infusion, so a rate and a clearance both have to be assembled from what the stem gives

The infusion stops after a stated time rather than at steady state, so the plateau is scaled by the fraction reached. The stem gives a total dose with a duration and a rate constant with a volume, not a rate and a clearance, so those are built first.

ALGEBRA

$$R = \frac{150 \text{ mg}}{6 \text{ hr}} = 25 \text{ mg/hr}$$

The stem gives a total amount and a duration rather than a rate, so the rate has to be formed first. Milligrams divided by hours gives milligrams per hour, which is the unit R must carry.

ALGEBRA

$$Cl = k \cdot V_D = (0.231 \text{ hr}^{-1})(15 \text{ L}) = 3.465 \text{ L/hr}$$

Clearance is needed for the plateau and is the product of the rate constant and the volume of distribution, both of which are given. Reciprocal hours multiplied by litres leaves litres per hour.

ALGEBRA

$$C_{ss} = \frac{R}{Cl} = \frac{25 \text{ mg/hr}}{3.465 \text{ L/hr}} = 7.215 \text{ mg/L}$$

This is the concentration the infusion would reach if it ran indefinitely, and it is the height the 6-hour value is a fraction of. The hours cancel and milligrams per litre remain.

ALGEBRA

$$1 - e^{-(0.231 \text{ hr}^{-1})(6 \text{ hr})} = 1 - e^{-1.386} = 0.7499$$

Six hours at a rate constant of 0.231 per hour is exactly two half-lives, since 0.693 divided by 0.231 is 3 hours. Two half-lives close three quarters of the gap, which is why the bracket comes out at 0.7499.

ALGEBRA

C_p at 6 hr = (7.215 mg/L)(0.7499) = 5.4107 mg/L

Multiplying the plateau by the fraction achieved gives the concentration at the moment the infusion stops. Because six hours is exactly two half-lives here, the answer should be three quarters of 7.215, and 5.41 is exactly that, which is the arithmetic check worth making.

ROUND

C_p at 6 hr = 5.41 mg/L

Two decimal places is the precision she carries concentrations to in this module. Her cross-check in the lecture uses this same 5.41 mg/L as the starting point for the post-infusion decay, so rounding here is what the next question in the set inherits.

DOES THE NUMBER MAKE SENSE?

> $k = 0.231\ hr^{-1}$ is a 3-hour half-life, so 6 hours is two half-lives and the level reaches three quarters of the plateau 7.215 mg/L, about 5.41 mg/L; it must stay below 7.215.

Example 6 Her key

Answer: 2.71 mg/L

WHAT THE STEM GIVES

dose	150 mg	over the duration gives $R = \frac{150}{6} = 25$ mg/hr
duration	6 hours	the infusion time, in the C_{peak} exponent
k	$0.231\ hr^{-1}$	with V_D gives Cl = 3.465 L/hr; then the decline exponent kt $= 0.231 \times 3$
V_D	15 L	one factor of $Cl = k \cdot V_D$
t	3 hours	the time since cessation, in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

Two stages: build the concentration at the end of the 6-hour infusion, then decay it for 3 hours with $C_p = C_{peak} \cdot e^{-kt}$

The concentration is asked for after the infusion stopped, so no plateau can be read off and the problem splits in two. Stage one uses a rate and a clearance to find how high the curve rose; stage two decays that height first order, with the 3 hours counted from cessation, not from the start.

ALGEBRA

$$R = \frac{150\ mg}{6\ hr} = 25\ mg/hr;\ Cl = (0.231\ hr^{-1})(15\ L) = 3.465\ L/hr$$

The rate comes from the amount and the duration, and the clearance from the rate constant and the volume of distribution. Both are needed before the concentration at the end of the infusion can be found.

ALGEBRA

$$C_{peak} = \frac{25\ mg/hr}{3.465\ L/hr}(1 - e^{-(0.231)(6)}) = (7.215\ mg/L)(0.7499) = 5.41\ mg/L$$

The infusion ran for only two half-lives, so it reached 75 per cent of its plateau rather than the plateau itself. This is the concentration at the moment the infusion stops, and it is what the decay starts from.

SETUP

$C_p = C_{peak} \cdot e^{-kt}$, t measured from cessation

With the input switched off only first-order elimination remains, so the post-infusion curve is the same decay used after a bolus. The 3 hours is counted from the end of the infusion, not from its start.

ALGEBRA

$$C_p = (5.41\ \text{mg/L})e^{-(0.231\ \text{hr}^{-1})(3\ \text{hr})} = (5.41\ \text{mg/L})(0.5) = 2.705\ \text{mg/L}$$

Three hours is exactly one half-life for this drug, since 0.693 divided by 0.231 is 3, so the concentration halves and the exponential factor is 0.5. That is the cross-check she uses in the lecture, and it avoids the calculator altogether.

ROUND

$$C_p = 2.71\ \text{mg/L}$$

Two decimal places follows the precision of the 5.41 mg/L the decay started from. 2.7 mg/L, the same number to one decimal place, also carries full marks on her paper.

DOES THE NUMBER MAKE SENSE?

The end-of-infusion level is 5.41 mg/L, and $k = 0.231\ \text{hr}^{-1}$ is a 3-hour half-life, so 3 hours after stopping exactly half remains, 2.71 mg/L; the answer must be below 5.41.